PYTHON FOR DATA SCIENCE · LESSON 19 OF 29

pandas IV: Dates, Time Series, Resampling and Rolling Windows

Work with time. Parse dates, extract parts with .dt, do date arithmetic, set a DatetimeIndex, resample to weeks, months and quarters, compute rolling and expanding windows, shifts and growth rates, and build a cohort-style monthly report with the daily sales data.

Python class 15 | pandas 2.3 / 3.0 | Intermediate | 110 min

By the end of this lesson you can

- Parse and format dates and extract year, month, weekday and quarter
- Do date arithmetic with Timedelta and DateOffset, including business days
- Resample time series and use rolling, expanding and exponentially weighted windows
- Compute period-over-period change, year-over-year growth and cumulative totals with shift, pct_change and cumsum
- Handle time zones and missing dates

Datasets used

`daily_sales.csv` `superstore_sales.csv`

Parsing dates

Python

```
import pandas as pd, numpy as np
pd.set_option("display.max_columns", 20); pd.set_option("display.width", 160)
d = pd.to_datetime(pd.Series(["2025-01-15", "2025-03-01", "2025-12-31"]))
print(d.dt.year.tolist(), d.dt.month_name().tolist(), d.dt.day_name().tolist(),
d.dt.quarter.tolist(), d.dt.isocalendar().week.tolist())
print(pd.to_datetime("05/01/2025", dayfirst=True).date(),
pd.to_datetime("05/01/2025").date())     # 5 Jan vs 1 May!
print(pd.to_datetime("2025-01-15 14:30").strftime("%d %b %Y, %I:%M %p"),
pd.Timestamp("2025-02-28").is_month_end)
```

OUTPUT

```
[2025, 2025, 2025] ['January', 'March', 'December'] ['Wednesday', 'Saturday',
'Wednesday'] [1, 1, 4] [3, 9, 1]
2025-01-05 2025-05-01
15 Jan 2025, 02:30 PM True
```

WATCH OUT **Day-first dates**

Indian and European data often write 05/01/2025 for 5 January. Pass `dayfirst=True` or, better, an explicit `format="%d/%m/%Y"`. Check a few parsed values every time.

`.dt` accessor	Gives
`.year`, `.month`, `.day`, `.hour`	Parts
`.day_name()`, `.month_name()`	Text names
`.dayofweek` (Mon = 0), `.quarter`, `.isocalendar().week`	Calendar positions
`.to_period("M")`, `.to_period("Q")`	Month / quarter periods
`.floor("D")`, `.normalize()`	Truncate time
`.is_month_end`, `.days_in_month`	Calendar flags

Date arithmetic

Python

```python
s = pd.read_csv("superstore_sales.csv", parse_dates=["order_date", "ship_date"])
s["ship_days"] = (s["ship_date"] - s["order_date"]).dt.days
print(s.groupby("ship_mode")["ship_days"].agg(["mean", "max"]).round(1))
t = pd.Timestamp("2025-01-31")
print(t + pd.Timedelta(days=30), t + pd.DateOffset(months=1), t + pd.offsets.MonthEnd(1), t
+ pd.offsets.BDay(3))
print(np.busday_count("2025-10-01", "2025-11-01"), "working days in October 2025 (Mon-
Fri)")
print(pd.date_range("2025-01-01", periods=4, freq="W-MON").date.tolist(),
pd.date_range("2025-01-01", "2025-06-30", freq="QE").date.tolist())
```

OUTPUT

```
                mean  max
ship_mode
First Class      1.5    2
Same Day         0.0    0
Second Class     3.0    4
```

```
Standard Class   5.6    7
2025-03-02 00:00:00 2025-02-28 00:00:00 2025-02-28 00:00:00 2025-02-05 00:00:00
23 working days in October 2025 (Mon-Fri)
[datetime.date(2025, 1, 6), datetime.date(2025, 1, 13), datetime.date(2025, 1, 20),
datetime.date(2025, 1, 27)] [datetime.date(2025, 3, 31), datetime.date(2025, 6, 30)]
```

`Timedelta` adds exact durations; `DateOffset` and offsets follow the calendar (month ends, business days).

A time series with a DatetimeIndex

Python

```python
daily = pd.read_csv("daily_sales.csv", parse_dates=["date"]).set_index("date").sort_index()
print(daily.shape, daily.index.min().date(), "to", daily.index.max().date())
print(daily.loc["2023-03"].head(3))                    # partial-string indexing: all of
March 2023
print(daily.loc["2023-12-24":"2023-12-27"])
full = pd.date_range(daily.index.min(), daily.index.max(), freq="D")
print("missing days:", len(full.difference(daily.index)))
```

OUTPUT

```
(1461, 1) 2022-01-01 to 2025-12-31
            units_sold
date
2023-03-01         260
2023-03-02         272
2023-03-03         234
            units_sold
date
2023-12-24         343
2023-12-25         270
2023-12-26         232
2023-12-27         257
missing days: 0
```

Resampling

`resample` is `groupby` for time: change the frequency and aggregate.

Python

```
monthly = daily["units_sold"].resample("ME").sum()
print(monthly.head(4))
q = daily["units_sold"].resample("QE").agg(["sum", "mean", "max"]).round(1)
print(q.tail(4))
wk = daily["units_sold"].resample("W-SUN").sum()
print(wk.describe().round(0))
print(daily.groupby(daily.index.day_name())
["units_sold"].mean().round(1).sort_values(ascending=False))
```

OUTPUT

```
date
2022-01-31    5998
2022-02-28    5797
2022-03-31    6756
2022-04-30    7156
Freq: ME, Name: units_sold, dtype: int64
              sum   mean  max
date
2025-03-31  30462  338.5  405
2025-06-30  34592  380.1  440
2025-09-30  34786  378.1  432
2025-12-31  33080  359.6  415
count     210.0
mean     2073.0
std       391.0
min       420.0
25%      1766.0
50%      2073.0
75%      2362.0
max      2801.0
Name: units_sold, dtype: float64
date
Sunday       323.3
Saturday     320.4
Friday       289.5
Tuesday      288.6
Thursday     288.5
Monday       287.9
Wednesday    287.3
Name: units_sold, dtype: float64
```

Frequencies: `D` day, `W` week, `ME` month end, `MS` month start, `QE` quarter, `YE` year, `h` hour, `min` minute.

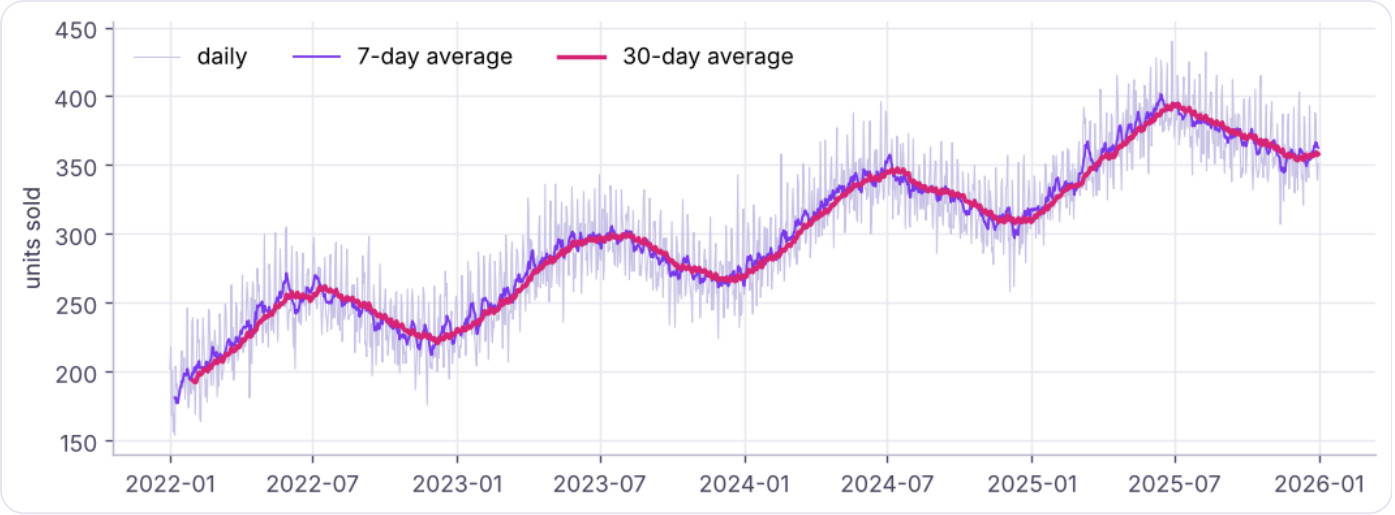
Rolling and expanding windows

Python

```
ts = daily["units_sold"]
df = pd.DataFrame({"units": ts, "ma_7": ts.rolling(7).mean(), "ma_30":
ts.rolling(30).mean(), "ewm_14": ts.ewm(span=14).mean(),
                   "max_so_far": ts.expanding().max(), "cum_units": ts.cumsum()})
print(df.iloc[[0, 6, 29, 400, -1]].round(1))
```

OUTPUT

date	units	ma_7	ma_30	ewm_14	max_so_far	cum_units
2022-01-01	202	NaN	NaN	202.0	202.0	202
2022-01-07	154	181.1	NaN	176.0	218.0	1268
2022-01-30	239	198.0	193.7	203.2	246.0	5811
2023-02-05	273	252.6	244.1	251.9	305.0	93681
2025-12-31	351	362.6	357.9	359.0	440.0	435285



Daily units with 7-day and 30-day moving averages: averages smooth out daily noise and reveal trend and seasonality.

Shifts and growth

Python

```python
m = monthly.to_frame("units")
m["prev_month"] = m["units"].shift(1)
m["mom_%"] = (100 * m["units"].pct_change()).round(1)
m["yoy_%"] = (100 * m["units"].pct_change(12)).round(1)          # same month last year
m["ytd"] = m.groupby(m.index.year)["units"].cumsum()
print(m.tail(4))
yr = daily["units_sold"].resample("YE").sum()
print(yr, (100 * yr.pct_change()).round(1).tolist())
```

OUTPUT

```
            units  prev_month  mom_%  yoy_%     ytd
date
2025-09-30  11071     11798.0   -6.2   12.3   99840
2025-10-31  11291     11071.0    2.0   15.3  111131
2025-11-30  10721     11291.0   -5.0   14.7  121852
2025-12-31  11068     10721.0    3.2   15.1  132920
date
2022-12-31     85031
2023-12-31    100536
2024-12-31    116798
2025-12-31    132920
Freq: YE-DEC, Name: units_sold, dtype: int64 [nan, 18.2, 16.2, 13.8]
```

`shift(1)` is SQL's `LAG`; `shift(-1)` is `LEAD`; `pct_change(12)` on monthly data is year-over-year growth.

Time zones

Python

```python
ts_utc = pd.Timestamp("2025-03-10 09:00", tz="UTC")
print(ts_utc.tz_convert("Asia/Kolkata"), ts_utc.tz_convert("America/New_York"))
naive = pd.Series(pd.to_datetime(["2025-03-10 09:00"]))
print(naive.dt.tz_localize("Asia/Kolkata").dt.tz_convert("UTC").tolist())
```

OUTPUT

```
2025-03-10 14:30:00+05:30 2025-03-10 05:00:00-04:00
[Timestamp('2025-03-10 03:30:00+0000', tz='UTC')]
```

Store timestamps in UTC; convert to local time only for reporting.

Mini case: monthly KPI report

Python

```
s["month"] = s["order_date"].dt.to_period("M")
kpi = s.groupby("month").agg(sales=("sales", "sum"), orders=("order_id", "nunique"),
customers=("customer_id", "nunique"))
kpi["aov"] = (kpi.sales / kpi.orders).round(0)
kpi["sales_3m_avg"] = kpi.sales.rolling(3).mean().round(0)
kpi["yoy_%"] = (100 * kpi.sales.pct_change(12)).round(1)
print((kpi.tail(6).assign(sales=lambda d: (d.sales / 1e6).round(2), sales_3m_avg=lambda d:
(d.sales_3m_avg / 1e6).round(2))))
first = s.groupby("customer_id")
["order_date"].min().dt.to_period("M").rename("first_month")
s2 = s.join(first, on="customer_id")
new_vs_repeat = s2.assign(kind=np.where(s2.month == s2.first_month, "new",
"repeat")).groupby(["month", "kind"])["customer_id"].nunique().unstack().tail(3)
print(new_vs_repeat)
```

OUTPUT

```
         sales  orders  customers      aov  sales_3m_avg  yoy_%
month
2025-07   7.19      99         91  72627.0          5.86   49.2
2025-08   5.80      99         91  58608.0          5.65    0.8
2025-09   8.47     107        101  79141.0          7.15   58.9
2025-10   6.14     124        115  49518.0          6.80  -15.4
2025-11  10.28     152        139  67661.0          8.30   41.2
2025-12   7.65     134        121  57106.0          8.03   15.3
kind     new  repeat
month
2025-10  8.0   107.0
2025-11  4.0   135.0
2025-12  5.0   116.0
```

Summary

KEY TAKEAWAYS

- Parse with `pd.to_datetime` (give `format` or `dayfirst`); use `.dt` for parts and `to_period` for months/quarters.
- `Timedelta` = exact duration; `DateOffset` / `MonthEnd` / `BDay` = calendar-aware moves; `date_range` builds calendars.
- A sorted `DatetimeIndex` allows partial-string slicing (`df.loc["2025-03"]`) and `resample`.
- Rolling, expanding and EWM windows smooth series; `shift` = lag/lead, `pct_change(n)` = growth, `cumsum` = running totals.
- Keep timestamps in UTC, check for missing dates, and convert for display.

Quick quiz

6 questions · tap an answer to check it

1. What does `daily.resample("ME").sum()` produce?

A Daily totals

B Monthly totals (labelled at month end)

C A 7-day average

D An error

Answer: B. ME = month end frequency.

2. Which gives year-over-year growth on monthly data?

A `pct_change(1)`

B `pct_change(12)`

C `shift(-12)`

D `rolling(12).mean()`

Answer: B. Compare each month with the same month a year earlier.

3. `shift(1)` is equivalent to which SQL function?

A LAG

B LEAD

C ROW_NUMBER

D NTILE

Answer: A. `shift(-1)` is LEAD.

4. How do you parse "05/01/2025" as 5 January?

A `pd.to_datetime(x)`

B `pd.to_datetime(x, dayfirst=True)` or `format="%d/%m/%Y"`

C `.dt.day`

D `astype(str)`

Answer: B. The default assumes month first.

5. What does `ts.rolling(7).mean()` give for the first 6 rows?

A 0

B NaN (not enough values yet)

C The value itself

D An error

Answer: B. Use `min_periods=1` to allow partial windows.

6. What is the difference between `Timedelta(days=30)` and `DateOffset(months=1)` from 31 Jan?

A Timedelta adds exactly 30 days (2 Mar); DateOffset moves to the same day next month, clipped to 28 Feb

B They are identical

C Timedelta is calendar-aware

D DateOffset adds 30 days

Answer: A. Use calendar offsets for month logic.

EXERCISE

Time-series practice (daily_sales.csv)

(1) Which month had the highest total units? (2) Compute the 28-day rolling sum and its maximum. (3) Average units by weekday and month (a heatmap-ready pivot). (4) Year-over-year growth by quarter. (5) Flag days where units exceed the 30-day average by 50%.

Show solution (worked answer)

Python

```
print(monthly.idxmax(), monthly.max())
print(ts.rolling(28).sum().max())
pv = daily.assign(wd=daily.index.day_name(),
m=daily.index.month).pivot_table(index="wd", columns="m", values="units_sold",
aggfunc="mean").round(0)
qq = ts.resample("QE").sum(); print((100 * qq.pct_change(4)).round(1).dropna())
spikes = daily[ts > 1.5 * ts.rolling(30).mean()]; print(len(spikes))
```

INTERVIEW CORNER

Questions you may be asked

Q: resample vs groupby? `resample` is a time-based groupby on a DatetimeIndex that also creates empty periods; `groupby` with `dt.to_period` is similar but only returns periods present in the data.

Q: How do you compute a 7-day moving average in pandas and in SQL? `s.rolling(7).mean()` ; in SQL `AVG(x) OVER (ORDER BY d ROWS BETWEEN 6 PRECEDING AND CURRENT ROW)` .

Q: How do you handle missing dates in a daily series? Reindex to a full `date_range` , then decide: fill with 0 (no sales), forward-fill (stock levels), or interpolate (sensor readings).

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.